SinMULU-JR at the Sinhala MMLU Shared Task (ICTer 2026): English Reasoning over Sinhala Questions with Gemma-3-4B

Technical report: data, method, every experiment, and the first submission

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Abstract

We describe our system and the complete experimental record for the Sinhala MMLU Shared Task at ICTer 2026, whose test set consists of 2,238 four-option multiple-choice questions from Sri Lankan 2025 school examinations (Easy, Medium, Hard). Under the task's 8B-parameter, open-weight, no-retrieval rules and a time budget of a few days, we use a single model, `google/gemma-3-4b-it` (4.30B parameters), without fine-tuning, and study 59 inference-time conditions: prompt wording, subject and role information, few-shot examples, option-order averaging, contextual calibration, cloze scoring, 4-bit quantisation, a pretrained base model, chain-of-thought reasoning in Sinhala and in English with different length limits, self-consistency, prompt ensembles, three Sinhala→English translation pipelines (NLLB-200, MADLAD-400, self-translation) with direct and reasoning-based answering, prompts that target negation and option-by-option judgement, a three-turn negation-consistency chain, and option-order averaging of the reasoning. Conditions were screened on a 280-question stratified subset of the public (Easy-only) training data and the best were confirmed on 1,483 held-out questions under selection rules fixed before the confirmation results were seen. **Letting Gemma reason step by step in English about the Sinhala question, then scoring the answer digit, raises accuracy from 38.2% to 45.6% (+7.4 points, Holm-corrected $p = 1.2 \times 10^{-6}$, $n = 1{,}483$).** Translation pipelines, calibration, option-order averaging, cloze scoring and self-consistency did not help. Translations drop the negation of about 40% of negated questions, which explains why translated input hurts most on exactly those questions, whereas English reasoning over the original Sinhala handles them well. We also report a scoring pitfall that silently reduces next-token evaluation of Gemma to chance: the space before the answer digit is a separate token, so the reply must be pre-filled with "පිළිතුර: " (with a trailing space); without it only 5% of the probability falls on the four digits. Two submissions have been produced and validated: English reasoning with a teacher role, limited to 512 and to 1,024 new tokens. The final system was reproduced from the committed code on a different machine with identical answers for all 2,238 test questions. Leaderboard scores: [to be filled in after the Codabench upload].

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1 Introduction

SinhalaMMLU [1] is the first multitask language-understanding benchmark for Sinhala, built from Sri Lankan national school curricula and examinations, with questions grouped by difficulty: Easy (grades 6–8), Medium (O/L level) and Hard (A/L level). The paper finds that open models up to 8B parameters are close to random (25%) and that even Qwen2.5-7B-chat reaches only 30.85% on Easy questions zero-shot (Table 15 of the paper), while Claude 3.5 Sonnet reaches 67.65% overall. The ICTer 2026 shared task asks participants to improve on this under strict constraints (Section 2).

Our constraints were mainly time and compute: work on the final system started on 24 October, three days before the leaderboard closed, on a single rented RTX 3090. We therefore chose one strong small multilingual model and investigated only inference-time methods, with a statistically careful protocol so that the many comparisons would not mislead us.

Contributions.

1. A reproducible evaluation pipeline for Sinhala MCQs (`sinmmlu/`) with label-mass validity checks, paired significance tests, and a strict separation of screening, confirmation and test data.
2. A systematic screen of 59 inference-time conditions for Gemma-3-4B-it on Sinhala exam questions, and pre-registered confirmations of the best candidates on 1,483 held-out questions.
3. A verified reproduction: rebuilt from the committed code on a second machine, the final system gives the same answer on every one of the 2,238 test questions.
4. The finding that *English* chain-of-thought over the untranslated Sinhala question gives a robust gain of about +7 points, while machine translation (NLLB, MADLAD, self-translation) does not help, also not when combined with reasoning, because translations frequently lose negation.
5. A documented scoring pitfall (answer pre-fill and the space token) and several other engineering errors that we found and corrected, with their effect on results.

2 The shared task

2.1 Rules

Rule	How our system complies
$\leq$ 8B total parameters across all inference components (MoE counts all experts, LoRA adapters add to the base, quantisation does not reduce the count)	One model: Gemma-3-4B-it, 4,300,079,472 parameters including its unused vision tower (Section 4.2).
Open-weight models released before 1 October 2026	Gemma 3 was released in March 2025 (Gemma licence).
No retrieval of any kind (RAG, BM25, dictionaries, lookup tables)	None. Few-shot examples are *fixed per subject* and never chosen by similarity to the test question.
Closed APIs only for generating training data, with disclosure	No closed API was used at all.
No training, tuning or prompt design on test questions	Every choice was made on the public training data; test questions were only scored to write predictions (one process note in Section 12).
Organisers reproduce every final system from submitted code	Deterministic greedy decoding, pinned library versions, model revision hash, one command per step (Section 11).
Cite Pramodya et al., EMNLP 2025	[1].

Table 1: Task rules and compliance.

2.2 Metric, submission format and schedule

The leaderboard metric is the **macro average of the Easy, Medium and Hard accuracies**. Submissions are made on Codabench (at most 3 per day) as a zip containing `predictions.jsonl`, one line per test question, e.g. `{"q_no": 1, "subject": "Art", "category": "Humanities", "level": "easy", "answer": "2"}`. The `answer` is a string; the key fields must be copied verbatim from the test files (their subject strings are inconsistent, Section 3). Schedule: leaderboard closes 27 October 2026 (23:59), code due 28 October, system report and slides due 30 October.

3 Data

3.1 Public training and test sets

Set	Questions	Files (subjects)	Options	Years / level
Public train (with answers)	1,851	14	4	2018–2023, Easy only (grades 6–8)
Public test, Easy	642	13	4	2025
Public test, Medium	1,111	13	4	2025
Public test, Hard	485	15	4	2025
Public test, total	2,238	41	4	no answers

Table 2: Public data provided by the organisers.

Training-set subjects. Arts 100, Buddhism 162, Catholicism 132, Christianity 130, Civics 123, Eastern music 143, Geography 117, Health and Physical Science 130, History 151, Islam 108, Sinhala language and literature 155, Dancing 108, Drama and theatre 128, Science 164. Answer distribution: 1: 473, 2: 501, 3: 507, 4: 370 (option 4 is under-represented, 20.0%).

Test-set subjects. Easy: the same 13 school subjects as above minus Islam (Art 30, Buddhism 30, Catholicism 50, Christianity 58, Civics 30, Dancing 39, Drama 60, Eastern Music 44, Geography 50, Health 36, History 40, Science 92, Sinhala 83). Medium (O/L): Agriculture and Food Technology 37, Buddhism 174, Catholicism 40, Christianity 38, Civics 159, Communication and Media Studies 40, Dancing 153, Design and Construction Technology 37, Eastern Music 36, Geography 41, Health 72, History 125, Sinhala 159. Hard (A/L): Agricultural Science 37, Art 15, Biology 50, Biosystem Technology 41, Buddhist Civilization 40, Business Studies 29, Chemistry 14, Communication and Media Studies 50, Dancing (Indigenous) 22, Economics 46, Geography 40, Home Economics 29, Physics 22, Political Science 30, Sinhala 20. Many Medium and Hard subjects (e.g. Biology, Economics, Political Science) have no training data at all.

Data issues found and handled.

- **Label error:** Geography (training file `Geography_social_science_easy_f_117.json`) question 64 had an invalid answer; it was corrected to 3 (2026-10-14).
- **Inconsistent subject strings** inside one file, e.g. "Eastern_Music" and "Eastern Music", "drama and Theatre " (trailing space), and different English names between train and test ("Arts"/"Art", "Health and Physical Science"/"Health"). Submission keys are copied verbatim; internally every question gets a unique id `<level>/<file>/q<q_no>`, and the source file is used as the subject key.
- **Zero-width characters:** the zero-width non-joiner (U+200C) appears in 357 of 2,238 test items but only 29 of 1,851 training items (a legacy-font artefact); it is removed from model inputs. The zero-width joiner (U+200D) is kept because it is part of Sinhala conjuncts such as ශ්‍රී.

Data we did not use. The Hugging Face release `naist-nlp/SinhalaMMLU` was *deliberately not accessed*: its card states that the shared-task Hard test set ("SinhalaMMLU 2.0") is built from the paper's Hard questions with one option removed, so it may contain test questions with answers. It is also gated under CC BY-NC-ND 4.0. No other external data was used for the system described here.

3.2 Contamination check

Every training question was compared with every test question by character 5-gram Jaccard similarity on a normalised text (NFC, ZWJ/ZWNJ removed, punctuation and symbols mapped to spaces, lower-cased Latin, blank-runs such as "…" and "_____" removed), computed two ways and exactly for every candidate pair: *full* = question + options sorted alphabetically (years reorder options), and q = question only (only for questions of ≥ 25 characters). Classes: remove if full ≥ 0.5; review if full is in $[0.35, 0.5)$ or ($q \geq 0.5$ and full ≥ 0.2); ignored if $q \geq 0.5$ and full < 0.2 (generic stems such as "choose the correctly spelled word"). Stricter thresholds (0.35/0.25) are defined for collected past papers.

Candidates vs. public test (2,238)	remove	review	ignored
Public train (1,851)	18 pairs / 18 items	100 pairs	21 pairs

Table 3: Near-duplicates between public train and test. The 18 remove items (full Jaccard 0.5–0.83) are excluded from every evaluation set used for decisions (`easy_clean`).

3.3 Evaluation sets

Because the team decided to keep the public training data for *evaluation only*, all model selection uses it as a development set. All sets are built deterministically by `python -m sinmmlu.build_dev` (seed 42); their id lists are version-controlled, and rebuilding on the pod gave byte-identical id lists (MD5 checked).

File	n	Content and use
`easy_full`	1,851	all public train
`easy_clean`	1,833	minus the 18 test near-duplicates
`inference_subset`	280	**screening set**: 20 random questions from each of the 14 subjects of `easy_clean` (seed 42). Gold answers 1/2/3/4 = 77/74/74/55, so always answering "1" gives 27.5% (the chance line); 95% interval about ±5.6 points.
`fewshot_pool`	70	5 fixed examples per subject, not in the subset (seed 43). The first four of each subject have four different answers, so that a k-shot prompt does not teach a position bias. The only source of few-shot examples.
`inference_rest`	1,483	**confirmation set**: the rest of `easy_clean`; disjoint from the two above; 95% interval about ±2.5 points.

Table 4: Evaluation sets (all from public train, all Easy).

4 Model choice

4.1 Phase 0: small models

Before the final week, the team screened three small instruction-tuned models on all 1,851 Easy questions with next-token scoring of the digits 1–4 (Table 5). All were at chance. Tokeniser *fertility* (tokens per whitespace-separated Sinhala word, over all 1,851 questions) strongly favoured Gemma: 2.96 for Gemma-3-1B-it, against 8.66 for Qwen3-0.6B and 10.74 for Llama-3.2-1B-Instruct. Gemma's prompts are therefore about three times shorter, and its vocabulary covers Sinhala far better.

Model (chat template on)	1 order	4 orders	calibrated	label mass
Qwen3-0.6B ("strict" prompt)	25.6	24.9	26.6	0.010
Gemma-3-1B-it	26.6	28.3	26.7	0.984
Llama-3.2-1B-Instruct	25.6	23.7	26.5	0.873

Table 5: Phase 0 (14–15 October, Colab T4, full precision), accuracy (%) on `easy_full` (n=1,851, ±2.0 points). "Label mass" is the total probability on the four digit tokens. With the chat template off, every model had a label mass of 0.000; Section 5.2 explains why.

4.2 Gemma-3-4B-it

Given the time available, we chose `google/gemma-3-4b-it`[1] (context length 131,072 tokens), loaded in bfloat16. It is a multimodal checkpoint; we use text input only, but count all its parameters. Parameters counted from the stored safetensors: **4,300,079,472** = 3,880,263,168 (language model) + 419,816,304 (vision tower and projector).

[1] Hugging Face revision `093f9f388b31de276ce2de164bdc2081324b9767`.

Correction. Our first parameter counter summed the parameters of an un-initialised (meta-device) model without tying shared matrices and therefore counted tied embeddings twice: it reported 4.97B for Gemma-3-4B, 3.46B for MADLAD-400-3B and 1.40B for NLLB-600M. The correct values are 4.30B, 2.94B (MADLAD stores an untied output layer) and 0.62B. An earlier statement in our notes that Gemma + MADLAD ("8.44B") exceeds the limit was therefore wrong: the combination has 7.24B parameters and is allowed. This did not affect any decision (Section 7.2).

5 Method

5.1 Label scoring

For a question x with options $o_1, \ldots, o_4$, a prompt $\pi(x)$ is rendered (with Gemma's chat template unless stated otherwise), and the model's next-token distribution is read at the answer position. For each label $\ell \in \{1, 2, 3, 4\}$ we take $\log P(\ell \mid \pi(x))$ over the *full* vocabulary, renormalise over the four labels, and predict the argmax. This is the scoring used by the SinhalaMMLU paper for open models. We record the **label mass** $\sum_\ell P(\ell \mid \pi(x))$ for every question: if it is low, the model wanted to say something else, and the argmax over the four labels is weak evidence. A condition is only trusted if its mean label mass is ≥ 0.5. Prompts are never truncated (an over-long prompt raises an error), batches are left-padded with explicit position ids, and any non-finite logit stops the run.

5.2 The answer pre-fill and the space token

The first test run of the pipeline on Gemma-3-4B-it showed a mean label mass of only about 0.05. Inspecting the top next tokens revealed two causes. (i) Gemma begins its reply with a phrase such as "වැඩිම ගැළපෙන පිළිතුර: 2 …" ("the most suitable answer: 2") or with "**" (Markdown bold), so the digit is not the first token. (ii) Gemma's tokenizer encodes "␣2" as two tokens, a space and "2", and has no single token for the digit with a leading space, so after "…:" the model's next token is the space. Table 6 shows the effect on 40 questions.

Rendering	Text appended after the prompt	Mean label mass
chat template	nothing (generation prompt only)	0.053
chat template	"පිළිතුර:"	0.000
chat template	**"පිළිතුර:␣"**	**1.000**
chat template	"පිළිතුර:␣**"	1.000
raw text (no template)	nothing (prompt ends with "පිළිතුර:")	0.001
raw text	**"␣"**	**0.999**
chat, English instructions	**"Answer:␣"**	**1.000**

Table 6: Answer pre-fill probe, Gemma-3-4B-it, first 40 questions of the subset. ␣ marks the trailing space.

All label-scored prompts are therefore scored with the reply pre-filled: in chat mode the model turn starts with "*answer-word*:␣" ("පිළිතුර" for Sinhala prompts, "Answer" for English ones), few-shot example answers are written the same way, and raw prompts get a trailing space. Only the bare-digit token is scored. With this, the label mass is ≥ 0.997 for every valid condition. The zero label mass of all chat-off runs in phase 0 (Table 5) is consistent with the same space-token effect. We did not test whether it affects other published next-token evaluations of Sinhala.

5.3 Option-order averaging and contextual calibration

Averaging scores each question under the 4 cyclic option orders $\sigma_s(i) = (i + s) \bmod 4$, maps the four renormalised label probabilities back to the original options, and averages them. **Contextual calibration** [2] estimates the model's label prior from three content-free inputs ("N/A", the empty string, and "[MASK]", each used as the question and as all four options), averages their log-probabilities, and subtracts this prior before renormalising. When the prompt shows the subject, one prior is estimated per subject.

5.4 Cloze scoring

The question is shown *without* options, and each option text o_i is scored as a continuation of the pre-filled answer: $\frac{1}{|o_i|} \sum_t \log P(o_{i,t} \mid \pi(x), o_{i,<t})$ (mean log-probability per token). The argmax over options is the

prediction.

5.5 Reasoning (chain of thought)

The model first generates a reasoning text with greedy decoding (at most 512 new tokens in English, 768 in Sinhala, since Sinhala needs about three times as many tokens per word; the follow-up experiments and submission 2 use other limits, Section 8). We then (i) find the last "Answer: *d*" or "පිළිතුර: *d*" it wrote (the *parse rate* is the fraction of questions where one is found), (ii) cut the reasoning just before it, (iii) append the cue "\n\n Answer:␣" (or "පිළිතුර:␣"), and (iv) score the four labels at that position as above. Every question therefore gets a probability vector, also when the reasoning was cut off by the length limit; where an answer was stated, the scored answer agrees with it in 99.6–100% of cases. Generation is deterministic (greedy, fixed seed 42), and the reasoning texts are cached so that an interrupted run resumes.

5.6 Few-shot examples

k-shot prompts ($k \in \{1, 3, 5\}$) put k solved examples *of the same subject* before the question, as earlier chat turns (user: the example question; assistant: "පිළිතුර: *d*"). The examples come only from the fixed `fewshot_pool`, in a fixed order, so every question of a subject sees the same examples; nothing is chosen by similarity to the question. Test subjects are matched to training subjects by the first three letters of the English subject name ("Art"↔"Arts", "Health"↔"Health and Physical Science", "Sinhala"↔"Sinhala language and literature"); subjects without training examples (most Medium and Hard subjects) use a fixed generic set (the first example of each subject).

Worked examples for reasoning. For `cot_en_fewshot2`, Gemma generated English reasoning (prompt `cot_en`) for the 70 pool questions; the 28 whose stated answer was correct were kept, up to and including their "Answer: *d*" line, as two worked examples per subject. Buddhism had no correct rationale and uses the generic fallback. These rationales are model-written, from public training data only.

5.7 Translation pipelines

The question and each option are translated separately into English and cached:

- **NLLB**: `facebook/nllb-200-distilled-600M` [3] (revision `f8d333a0…`, 615M parameters, CC-BY-NC 4.0), source `sin_Sinh`, target `eng_Latn`.
- **MADLAD**: `google/madlad400-3b-mt` [4] (revision `fa184c67…`, 2.94B parameters, Apache 2.0), input prefix "`<2en>`".
- **Self**: Gemma-3-4B-it itself, prompted "Translate the following Sinhala text into English. Write only the English translation, nothing else." (greedy, ≤ 512 new tokens).

NLLB and MADLAD are sentence-level systems: texts are split into lines and sentences (after ".", "?", "!"), each piece is translated with beam search (4 beams, ≤ 256 new tokens), and the pieces are re-joined; pieces without Sinhala letters are copied unchanged. Gemma then answers one of three prompt variants: English only, bilingual (Sinhala original followed by the English translation, with a note that the translation may contain errors), or the Sinhala question with English options. On the subset, NLLB translated 1,402 unique sentence pieces in about a minute; MADLAD translated 1,437 pieces (several minutes, run time not recorded).

5.8 Ensembles

An ensemble averages the four option probabilities of finished conditions of the *same* model (so the parameter count does not grow) and takes the argmax.

5.9 Prompts

All prompts are in Appendix A. The baseline `paper_nosubject` is Appendix B of the paper without the subject line.

6 Experimental protocol

Stages.

1. **Smoke test**: 8 questions through every code path (no results recorded). It found the pre-fill problem.

2. **Screening** on `inference_subset` (n=280) in three waves: (1) prompt wording, chat template, averaging, calibration, cloze, 4-bit, model references; (2) few-shot, NLLB and self-translation; (3) reasoning in Sinhala and English, MADLAD.
3. **Combinations** of the screening winners (also on the subset).
4. **Confirmation** on `inference_rest` (n=1,483) under a rule written into our lab notebook before the combination results were seen: take the four best *valid* conditions on the subset (label mass ≥ 0.5, parameter budget respected, ensembles only within one model) plus the baseline; a condition is *adopted* if it beats the baseline by ≥ 1 point with Holm-corrected exact McNemar $p < 0.05$ over the confirmed comparisons; the most accurate adopted condition wins; if none is adopted, the baseline is submitted.
5. **Test**: the winner is run once on the 2,238 test questions.

Statistics. Accuracy with a 95% Wilson interval; for every condition against the baseline on the same questions: the exact two-sided McNemar test on the discordant pairs, Holm–Bonferroni correction over all conditions compared in that table, and a paired bootstrap 95% interval of the accuracy gain (10,000 resamples of questions, seed 0).

Hardware and software. RunPod pod with one NVIDIA GeForce RTX 3090 (24 GB), 128 vCPUs, 251 GB RAM; Python 3.12.3, PyTorch 2.8.0 (CUDA 12.8), Transformers 5.18.0, Accelerate 1.15.0, bitsandbytes 0.50.2, PEFT 0.21.1. Gemma in bfloat16 with batch size 16 for scoring and generation; batches are split in halves automatically on GPU out-of-memory. 4-bit runs use NF4 with bfloat16 compute.

7 Results

7.1 Screening (280 questions)

Table 7 lists every condition. The baseline reaches 37.5%, already well above the 27.5% chance line.

Table 7: Screening on `inference_subset` (n=280; chance 27.5%). Gain = accuracy minus baseline (points); CI = paired bootstrap 95% interval of the gain; p = exact McNemar; p_{H} = Holm-corrected over the 39 comparisons of the final table; L = mean label mass. All with Gemma-3-4B-it, chat template, one option order, no calibration, bf16, unless stated.

Condition	Acc.	Gain	CI of gain	p	p_{H}	L
A: baseline						
A0 `paper_nosubject`	37.5	–	–	–	–	1.000
A0b same, chat template off	37.5	+0.0	[−4.3, +4.3]	1.00	1.00	0.997
B: prompt wording						
B1 `paper_subject_si` (Sinhala subject name)	42.1	+4.6	[+0.0, +9.6]	0.092	1.00	1.000
B2 `paper_subject_en` (English subject name)	40.7	+3.2	[−1.8, +8.2]	0.262	1.00	1.000
B3 `strict_nosubject` ("write only the digit")	36.8	−0.7	[−5.0, +3.2]	0.868	1.00	0.997
B4 `role_subject_si` (Sinhala teacher role + subject)	43.6	+6.1	[+1.1, +11.1]	0.030	0.87	1.000
B5 `en_instr_nosubject` (English instructions)	41.8	+4.3	[−0.4, +8.9]	0.088	1.00	1.000
B6 `letters_nosubject` (labels A–D)	36.4	invalid: label mass 0.002				0.002
B7 `fewshot1_nosubject`	40.7	+3.2	[−1.8, +8.6]	0.281	1.00	1.000
B7 `fewshot3_nosubject`	42.5	+5.0	[−0.7, +10.7]	0.120	1.00	1.000
B7 `fewshot5_nosubject`	42.1	+4.6	[−1.1, +10.4]	0.148	1.00	1.000
C: scoring and model						
C1 4 option orders	38.6	+1.1	[−3.6, +5.7]	0.766	1.00	1.000
C2 contextual calibration	34.3	−3.2	[−7.5, +0.7]	0.163	1.00	1.000
C3 4 orders + calibration	38.2	+0.7	[−4.3, +5.7]	0.888	1.00	1.000
C4 cloze (`cloze_nosubject`)	31.4	−6.1	[−13.6, +1.4]	0.128	1.00	–
C6 `cot_si` (reasoning in Sinhala)	43.2	+5.7	[−0.4, +11.8]	0.085	1.00	1.000
C7 `cot_en` (reasoning in English)	**47.9**	**+10.4**	[+4.3, +16.4]	0.0013	0.047	1.000
C8 4-bit NF4	36.8	−0.7	[−6.4, +5.0]	0.901	1.00	1.000
C9 Gemma-3-4B-**pt** (base model, raw prompt)	45.0	+7.5	[+0.7, +14.3]	0.046	1.00	0.917
C9 Gemma-3-1B-it	25.7	−11.8	[−19.3, −4.3]	0.0032	0.12	1.000
D: translation (Gemma answers; translator in brackets)						
D1 English only [self]	37.9	+0.4	[−6.4, +7.1]	1.00	1.00	1.000
D2 English only [NLLB]	37.1	−0.4	[−7.5, +6.8]	1.00	1.00	0.998
D3 English only [MADLAD]	43.9	+6.4	[−0.7, +13.6]	0.092	1.00	1.000
D4 bilingual [self]	37.5	+0.0	[−5.7, +5.7]	1.00	1.00	1.000
D4 bilingual [NLLB]	40.4	+2.9	[−2.9, +8.6]	0.382	1.00	1.000
D4 bilingual [MADLAD]	42.9	+5.4	[−0.4, +11.1]	0.082	1.00	0.998

Condition	Acc.	Gain	CI of gain	p	p_H	L
D5 Sinhala question, English options [self]	35.0	−2.5	[−8.9, +3.9]	0.510	1.00	0.997
D5 same [NLLB]	36.1	−1.4	[−8.2, +5.4]	0.755	1.00	1.000
D5 same [MADLAD]	37.9	+0.4	[−5.7, +6.4]	1.00	1.00	1.000
E: combinations						
`cot_en_fewshot2` (English reasoning + 2 worked examples)	**50.7**	**+13.2**	[+7.1, +19.3]	7.4e-5	0.003	0.996
`cot_en_role` (English reasoning + teacher role)	48.2	+10.7	[+3.9, +17.5]	0.0032	0.12	1.000
`cot_en_subject` (English reasoning + subject)	46.8	+9.3	[+2.9, +15.7]	0.0073	0.24	1.000
`fewshot3_subject_si`	42.9	+5.4	[−0.4, +11.1]	0.096	1.00	1.000
pt + `fewshot3_nosubject`	47.1	+9.6	[+3.6, +15.7]	0.0036	0.12	0.999
pt + `fewshot3_subject_si`	46.8	+9.3	[+3.2, +15.4]	0.0049	0.16	0.999
pt + `paper_subject_si`	40.4	+2.9	[−3.6, +9.6]	0.470	1.00	0.918
pt + `role_subject_si`	42.1	+4.6	[−2.1, +11.4]	0.218	1.00	0.894
ensemble `cot_en` + `cot_si`	49.3	+11.8	[+6.1, +17.9]	1.4e-4	0.005	–
ensemble `cot_en` + B4 + B7(3)	44.6	+7.1	[+2.1, +12.1]	0.0078	0.24	–
ensemble B4 + B7(3) + B1 + B5	44.3	+6.8	[+2.1, +11.4]	0.0079	0.24	–

Observations on the screening set.

- **Reasoning in English is the strongest single change** (+10.4, the only single condition significant after Holm). Reasoning in Sinhala helps less (+5.7) and is slower (450 s against 350 s for 280 questions).
- **Telling the model the subject** helps (+3 to +6), as the paper found; a Sinhala teacher role helps most. English instructions alone help about as much (+4.3).
- **Few-shot** examples help modestly (+3 to +5); 3 examples are as good as 5.
- **Option-order averaging** changes little (+1.1): Gemma has no strong position preference after the pre-fill fix. **Calibration hurts** (−3.2): it shifts predictions towards option 2 (46.8% of predictions, against 26.4% of gold answers), i.e. the content-free prior is a poor estimate of the model's bias here.
- **Cloze scoring is worse** (−6.1): Sinhala options of very different lengths and forms are not comparable by mean token log-probability.
- **4-bit NF4 costs nothing measurable** (−0.7, p=0.90), relevant if the organisers reproduce on a smaller GPU.
- **The pretrained base model** scores surprisingly well (45.0%) but has a strong preference for option 1 (46.4% of its predictions; 57.5% with the subject prompt) and a lower label mass (0.92), and it cannot follow the chat-style reasoning prompts that won.
- **Letters A–D** are not usable: Gemma answers with a Sinhala letter (e.g. ඊ) or a digit, so only 0.2% of the probability falls on the Latin letters.
- **NLLB and self-translation do not help** (between −2.5 and +2.9). MADLAD translation helps more (+6.4 English only, +5.4 bilingual) but stays below English reasoning; Gemma + MADLAD (7.24B) would be within the parameter limit.
- **Gemma-3-1B-it is at chance** (25.7%).

7.2 Selection

By the rule of Section 6, the confirmation candidates were the four best valid conditions on the subset: `cot_en_fewshot2` (50.7), the ensemble `cot_en`+`cot_si` (49.3), `cot_en_role` (48.2) and `cot_en` (47.9). At selection time MADLAD conditions were excluded as over budget because of the counting error of Section 4.2; with the correct count they would have been eligible, but the best MADLAD condition (43.9) ranked twelfth and would not have been selected.

7.3 Confirmation (1,483 held-out questions)

All four candidates beat the baseline clearly; the three English-reasoning variants lie within 0.8 points of each other, which is far inside their uncertainty, so the rule's tie-break (highest accuracy) chose `cot_en_role`. Two subset results did not replicate: the worked examples fell from 50.7% to 44.8%, and Sinhala reasoning from 43.2% (+5.7) to 36.8% (−1.3). This is the expected regression of the top of a 280-question ranking, and it is why decisions were taken only on the confirmation set.

Discordant pairs. Against the baseline, `cot_en_role` is right on 283 questions where the baseline is wrong and wrong on 173 where the baseline is right: reasoning changes many answers in both directions, with a clear net gain.

Condition	Acc.	Gain	CI of gain	p	p_{H}	Adopted	Parse rate	Time (s)
A0 baseline `paper_nosubject`	38.2	–	–	–	–	–	–	36
`cot_en_role`	**45.6**	**+7.4**	[+4.6, +10.3]	2.9e-7	1.2e-6	**yes, winner**	76.6%	2,084
`cot_en`	44.9	+6.7	[+3.9, +9.6]	5.5e-6	1.7e-5	yes	97.0%	1,871
`cot_en_fewshot2`	44.8	+6.6	[+3.7, +9.6]	1.6e-5	3.2e-5	yes	95.5%	2,791
ensemble `cot_en`+`cot_si`	44.0	+5.9	[+3.2, +8.6]	3.8e-5	3.8e-5	yes	–	–
`cot_si` (ensemble part only)	36.8	−1.3	[−4.0, +1.3]	0.36	–	–	97.4%	2,400

Table 8: Confirmation on `inference_rest` (n=1,483, ±2.5 points). p_{H} is Holm-corrected over the four candidates (the selection rule); `cot_si` was run only as a part of the ensemble and is not a candidate. Times are for the 1,483 questions on the RTX 3090.

7.4 Per-subject accuracy

Subject		n	A0	`cot_en`	`cot_en_role`	`cot_en_fs2`	`cot_si`
Arts	චිත්‍ර	70	42.9	44.3	45.7	42.9	50.0
Buddhism	බුද්ධ ධර්මය	137	33.6	44.5	40.2	40.9	36.5
Catholicism	කතෝලික ධර්මය	104	33.7	37.5	42.3	40.4	33.7
Christianity	ක්‍රිස්තියානි ධර්මය	103	38.8	51.5	49.5	52.4	32.0
Civics	පුරවැසි අධ්‍යාපනය	96	57.3	61.5	62.5	63.5	53.1
Eastern music	පෙරදිග සංගීතය	118	33.1	36.4	33.9	39.0	34.8
Geography	භූගෝල විද්‍යාව	90	34.4	52.2	40.0	50.0	41.1
Health and Phys. Sci.	සෞඛ්‍ය හා ශාරීරික විද්‍යාව	104	45.2	59.6	58.7	51.0	47.1
History	ඉතිහාසය	126	27.8	35.7	32.5	34.9	24.6
Islam	ඉස්ලාම් ධර්මය	83	41.0	50.6	59.0	51.8	36.1
Sinhala lang. and lit.	සිංහල භාෂාව සහ සාහිත්‍යය	130	35.4	33.1	40.0	34.6	29.2
Dancing	නර්තනය	82	40.2	26.8	40.2	32.9	23.2
Drama and theatre	නාට්‍ය හා රංග කලාව	101	35.6	46.5	37.6	42.6	38.6
Science	විද්‍යාව	139	42.4	51.8	60.4	54.0	41.7
All		1,483	38.2	44.9	45.6	44.8	36.8

Table 9: Accuracy (%) per subject on the confirmation set (descriptive only; per-subject intervals are ±8–12 points, so no decision was taken per subject).

English reasoning helps most on Geography (+17.8 for `cot_en`), Health (+14.4), Christianity (+12.7), Buddhism (+10.9), Drama (+10.9), Science (+9.4) and Islam (+9.6), i.e. content subjects. It does not help, or hurts, on the most language- and culture-bound subjects: Sinhala language and literature (−2.3) and Dancing (−13.4 for `cot_en`; 0.0 with the teacher role). This matches the paper's observation that culturally specific knowledge is the hardest part of SinhalaMMLU, and suggests that reasoning in English loses information when the question is about the Sinhala language itself.

7.5 Comparison with the paper

Our Easy-level numbers are on public-train questions (2018–2023) and the paper's on its own Easy split, so the comparison is indicative only. The paper reports zero-shot Easy accuracy of 30.85% for Qwen2.5-7B-chat, 34.31% for Llama-3.1-8B and 45.11% for Qwen2.5-72B-chat (Table 15 of [1]). Gemma-3-4B-it scored 38.2% with the paper's own prompt (pre-filled), and 45.6% with English reasoning, i.e. at the level the paper reports for a 72B model.

7.6 Test predictions: submission 1

`cot_en_role` was run on all 2,238 test questions in 3,083 s on the RTX 3090 (longest input 961 tokens). The file passed format validation twice (on the pod and locally): 2,238 lines, keys identical to the test set, every answer in {"1","2","3","4"}. The model stated an answer in only 62.4% of its reasoning texts (76.6% on the confirmation set), probably because the longer Medium and Hard questions more often reach the 512-token limit; every question still receives a scored answer, and the mean label mass after the cue was 0.998. This motivated the follow-up experiments below.

8 Follow-up experiments

All follow-up conditions were screened on the same 280 questions and compared with the winner at the same reasoning limit. Before each group was run, we fixed the rule that a condition goes to the confirmation set only if it beats that reference by at least 2 points on the subset.

8.1 Reasoning length, self-consistency and a prompt ensemble

Self-consistency [7] samples k reasoning texts per question (temperature 0.7, top-p 0.95, seeds $42 + 1000i$, reseeded per batch), scores each after the cue, and averages the k label distributions (soft vote); the hard majority vote of the stated answers is reported as well. The **ensemble** averages the probabilities of the three confirmed English-reasoning prompts.

Condition (subset, n=280)	Acc.	Stated answer	Note
`cot_en_role`, 512 new tokens (submission 1)	48.2	79.3%	about one answer in five cut off
`cot_en_role`, 768 new tokens	48.9	98.9%	
`cot_en_role`, 1,024 new tokens	48.9	100%	identical predictions to 768
`cot_en_role`, 1,536 new tokens	48.9	100%	identical
`cot_en_role`, 2,048 new tokens	48.9	100%	identical
`cot_en`, 1,024 new tokens	47.9	99.6%	same accuracy as at 512
`cot_en_role`, 1,024, self-consistency k=5	47.5	99.9%	majority vote also 47.5
ensemble `cot_en` + `cot_en_role` + `cot_en_fewshot2` (512)	50.4	–	

Table 10: Reasoning length, self-consistency and the three-prompt ensemble on the subset.

Gemma finishes every Easy reasoning text within about 768 tokens, so any limit from 768 upwards gives the same predictions; the limit only matters for answers that were cut at 512. Self-consistency is no better than greedy decoding at five times the cost and was not pursued. The ensemble was also computed on the confirmation set, where all its parts already existed: **46.7%** (+8.5 over the baseline, 95% CI [+5.8, +11.2]), but only +1.1 over `cot_en_role` (95% CI [−0.9, +3.0], $p = 0.31$). Because it was chosen after the confirmation results were known, this number is optimistic, and the ensemble was not adopted.

8.2 Translation combined with reasoning

The English-reasoning winner was given translated input: bilingual (the Sinhala question and options, then the machine translation, with a note that the Sinhala text is the original), English only, or a single prompt that asks Gemma to translate the question and options itself before reasoning (no second model).

Condition (subset, 1,024 tokens)	Acc.	Gain	CI of gain	p	Extra model
reference: `cot_en_role`, Sinhala input	48.9	–	–	–	none
`cot_en_role_translate_first`	50.4	+1.4	[−4.3, +7.1]	0.72	none
ensemble: Sinhala input + bilingual (MADLAD)	50.4	+1.4	[−3.2, +6.4]	0.67	MADLAD
`cot_en_role_bilingual`, MADLAD	49.3	+0.4	[−6.4, +6.8]	1.00	MADLAD (2.94B)
`cot_en_role_bilingual`, NLLB	47.9	−1.1	[−7.5, +5.0]	0.82	NLLB (0.62B)
`cot_en_role_en` (English only), MADLAD	44.3	−4.6	[−12.1, +2.5]	0.25	MADLAD
`cot_en_role_bilingual`, self-translation	41.8	−7.1	[−13.2, −0.7]	0.040	none

Table 11: Translation combined with English reasoning. No condition reached the pre-stated +2 points.

8.3 Negation and question types

For analysis only (never at inference), questions were tagged by simple patterns: *negation* (a Sinhala negation or "wrong" word such as නොවන, නොවේ, නොගැළපේ, වැරදි, අසත්‍ය, සාවද්‍ය in the question), *sub-option* (at least two options name a part letter A–E or a Roman numeral), *blank* ("…" or "_____"), *numeric* (a digit in the question or options), and *plain* (none). The subset has 34 negation, 5 sub-option, 16 blank, 30 numeric and 202 plain questions; the confirmation set 157, 22, 104, 119 and 1,123.

Translations lose negation. Of the 34 negated subset questions, the English translation still contains a negation ("not", "except", "incorrect", "false", …) in only 21 for MADLAD (62%), 20 for NLLB (59%) and 19 for self-translation (56%): roughly two negated questions in five become their opposite. On these

Condition (subset)	All	Negation (34)	Blank (16)	Numeric (30)	Plain (202)
A0 baseline (direct scoring)	37.5	35.3	25.0	33.3	39.1
`role_subject_si` (direct scoring)	43.6	38.2	56.3	43.3	43.1
`cot_si`	43.2	38.2	31.3	46.7	44.1
`cot_en_role`, 1,024	48.9	**58.8**	62.5	43.3	47.5
`cot_en_role_bilingual`, MADLAD	49.3	38.2	56.3	50.0	50.5
`cot_en_role_en`, MADLAD	44.3	29.4	–	–	–
`cot_en_role_negcheck`	51.4	50.0	43.8	46.7	52.5
`cot_en_role_each_option`	38.9	38.2	37.5	43.3	38.6

Table 12: Accuracy (%) by question type on the subset (descriptive; small groups). Sub-option (n=5) omitted. "–": not recorded before the raw outputs were lost (Section 12).

questions, English reasoning over the original Sinhala scores 58.8%, but 38.2% with bilingual MADLAD input and 29.4% with English-only input. With English reasoning, negated questions are *not* a weak spot (58.8% against 47.5% for plain questions); they are weak for direct scoring (35.3%) and for translated input. The groups are small, so these figures are indicative, but they explain why translation did not help.

Prompts for question types. Two reasoning prompts were tested: `cot_en_role_negcheck` ("Before reasoning, state whether the question asks for the correct option or for the option that is NOT correct, does NOT match, or is false.") and `cot_en_role_each_option` ("Evaluate each of the four options one by one and say whether it is correct or incorrect for this question."). On the subset, `negcheck` scored 51.4% (+2.5, 95% CI [−3.2, +8.2], $p = 0.48$) and met the +2 rule, although its gain came from non-negated questions (52.5% vs 47.5%) rather than negated ones (50.0% vs 58.8%). Judging each option hurt clearly: 38.9% (−10.0, CI [−15.7, −4.3], $p = 0.001$).

8.4 Second confirmation

Condition (confirmation set, n=1,483)	Acc.	Gain	p	p_{H}	Adopted
reference: `cot_en_role`, 512 tokens (submission 1)	45.6	–	–	–	–
`cot_en_role`, 1,024 tokens	45.6	±0.0	1.00	1.00	no (fallback)
`cot_en_role_negcheck`, 1,024 tokens	44.3	−1.3	0.33	0.67	no

Table 13: Second confirmation. Rule: adopt a candidate if it beats the reference by ≥ 1 point with Holm-corrected $p < 0.05$; otherwise use the 1,024-token version, which only removes cut-off answers.

The negation-check gain did not replicate. On Easy questions the 1,024-token limit gives the same accuracy as 512, since few Easy answers needed more; by the fallback rule it was used for submission 2 because it cannot truncate longer Medium and Hard reasoning.

8.5 Test predictions: submission 2

`cot_en_role` with a 1,024-token limit was run on the 2,238 test questions in 4,282 s (longest input 1,371 tokens). The model now states an answer in **99.1%** of its reasoning texts (62.4% for submission 1); mean label mass 0.998; format validated on the pod and locally. 140 of the 2,238 answers differ from submission 1; they are mostly questions whose reasoning was cut off at 512 tokens. No accuracy-related test statistic was computed.

Submission	Easy	Medium	Hard	Macro average
1: `cot_en_role`, 512 tokens	[–]	[–]	[–]	[–]
2: `cot_en_role`, 1,024 tokens	[–]	[–]	[–]	[–]

Table 14: Leaderboard results (to be filled in after the Codabench uploads).

8.6 Reproduction from a clean setup

On 26 October the final system was rebuilt on a different machine (a new RTX 3090 pod) from the committed code (`ae9abe5`): fresh library install, dev files rebuilt from the raw data (identical id lists), Gemma-3-4B-it

downloaded again (same revision `093f9f38…`), then the test run of submission 2 repeated (3,680 s; 4,282 s originally), recorded with the clean code revision `ae9abe5`. **All 2,238 test answers are identical to the uploaded submission 2.** The two confirmation-set runs repeated on the new machine also gave identical accuracies (baseline 38.17%, final system 45.58%), as did the final system on the 280-question subset (48.93%). Greedy decoding with bfloat16 on the same GPU type is therefore exactly reproducible with this code.

8.7 Question types on the confirmation set

The type analysis of Section 8.3 was repeated for the final system on the 1,483 confirmation questions, where the groups are larger (Table 15).

Condition (confirmation set)	All	Negation (157)	Sub-option (22)	Blank (104)	Numeric (119)	Plain (1,123)
A0 baseline (direct scoring)	38.2	37.6	27.3	28.8	35.3	39.2
final system (`cot_en_role`, 1,024)	**45.6**	**50.3**	31.8	35.6	37.0	46.3
gain	+7.4	**+12.7**	+4.5	+6.8	+1.7	+7.1

Table 15: Accuracy (%) by question type on the confirmation set (descriptive).

English reasoning helps negated questions most (+12.7), confirming the subset observation on five times as many questions. Sub-option (combinations of statements), fill-in-the-blank and numeric questions remain the weakest types (31.8–37.0%) and gain least: they are natural targets for later fine-tuning data.

8.8 Final screening round

Three more ideas were screened on the subset against the final system at 1,024 tokens (48.9%), with the same pre-stated rule (+2 points to go to the confirmation set):

- **Negation-consistency chain** (`cot_en_role_negchain`), a three-turn conversation: (1) reason in English and list *all* correct options ("more than one option may be correct", ≤ 1,024 tokens); (2) write the opposite question in English ("which options are NOT correct", or the reverse for an already negated question), reason and list its options (≤ 1,024 tokens); (3) with both analyses in the conversation, choose the single correct answer to the original question (≤ 256 tokens), scored after "Answer:␣".
- **Option-order averaging of the reasoning**: the final system reasons under the original option order and under the order shifted by one position (every option moves); the two scored label distributions are mapped back to the original options and averaged.
- **Three-prompt ensemble at 1,024 tokens**: `cot_en` + `cot_en_role` + `cot_en_fewshot2`, all at 1,024 tokens (the same 28 model-written worked examples as before).

Condition (subset, n=280)	Acc.	Gain	CI of gain	p	Time (s)
reference: `cot_en_role`, 1,024 tokens (final system)	48.9	–	–	–	438
negation-consistency chain	47.5	−1.4	[−7.9, +5.0]	0.74	1,236
option-order averaging of the reasoning (2 orders)	49.6	+0.7	[−3.9, +5.4]	0.88	857
ensemble of 3 reasoning prompts, 1,024 tokens	50.7	+1.8	[−2.9, +6.4]	0.54	1,273
part: `cot_en`, 1,024 tokens	47.9	−1.1	[−7.1, +5.0]	0.81	317
part: `cot_en_fewshot2`, 1,024 tokens	50.7	+1.8	[−4.3, +7.9]	0.65	517

Table 16: Final screening round. None reaches the pre-stated +2 points. Times on one RTX 3090 for 280 questions; the ensemble time is the sum of its three parts.

Why the negation chain does not help. The model followed the protocol (both lists were produced for 100% of the questions; a final answer was stated for 98.6%), but in step 1 it marked on average **2.0 options as correct** although exactly one is, and in step 2 it assigned 2.7 options to the opposite question; the final step inherits these inconsistent judgements. The chain hurts most on the questions it targets: negated questions fall from 58.8% to 47.1% and fill-in-the-blank questions from 62.5% to 43.8% (subset), presumably because writing the opposite of a "which is NOT …" question creates a double negation. Together with the earlier result for option-by-option judgement (−10.0), this indicates that Gemma-3-4B decides best when it reasons directly towards a single answer.

Averaging and ensembling. Averaging the reasoning over two option orders adds only +0.7 at twice the cost, so the reasoning has little position bias to cancel. The ensemble comes closest (+1.8) and is consistent with its 512-token result on the confirmation set (+1.1, $p = 0.31$): its real gain is probably about one point, below what the confirmation set can detect (±2.5 points) and at three times the test-time cost. Neither was confirmed; **submission 2 remains the final system.**

9 Final system

The current final system is submission 2; submission 1 differs only in the reasoning limit (512 tokens).

- **Model**: `google/gemma-3-4b-it`, revision `093f9f38…`, bfloat16, 4,300,079,472 parameters (only component; 8B limit respected).
- **Input**: NFC normalisation, ZWNJ removed, ZWJ kept; options numbered "1."–"4."; the English subject name from the test file.
- **Prompt** (user turn of Gemma's chat template):

```
You are an expert Sri Lankan teacher of {subject}.
The following is a multiple choice question in Sinhala about {subject}.
Think step by step in English, then choose the correct or most suitable answer from the options 1, 2, 3, 4.
End your reply with "Answer: <number>".

Question: {Sinhala question}
1. {option 1}
...
4. {option 4}
```

- **Decoding**: greedy, at most 1,024 new tokens (submission 2; 512 in submission 1); the stated answer (if any) is cut, "\n\n Answer:␣" appended, and the argmax over the digit tokens 1–4 is the prediction.
- **No** retrieval, no fine-tuning, no external data, no closed API, no test-time adaptation.

10 Engineering record: errors found and fixed

Small engineering errors can silently turn a benchmark run into noise. Table 17 lists every error we found, including our own.

Table 17: Errors found during the project.

Problem	Effect	Fix
YAML 1.1 parses unquoted `chat: on` as the boolean `True` (phase 0)	runs labelled "chat on" were run with the chat template off	chat mode as a command-line choice; old rows quarantined
Prompts truncated from the right at 1,024 tokens (phase 0)	the answer position was cut off for 9–78 Qwen/Llama prompts	no truncation; over-long prompts raise
Dedup "ignored" class depended on a thresholded zero (phase 0)	generic-stem pairs mislabelled	both Jaccards computed exactly for every pair
4-bit loading used bitsandbytes' default FP4 with fp16 compute; no NaN check	lower 4-bit quality; Gemma 3 can overflow in fp16	NF4 with bf16/fp32 compute; non-finite logits raise
The experiment grid forced full precision	4-bit runs impossible	`--quant` option
Calibration with the English subject prompt built its content-free inputs with the Sinhala subject name	wrong prior for that condition	content-free inputs copy the real row's metadata
No answer pre-fill; Gemma encodes the space before a digit as a separate token	label mass 0.05, accuracy close to random	pre-fill "*answer-word*:␣" (Section 5.2)
RunPod presets `HF_HOME` on the 30 GB volume	disk full during the MADLAD download	model cache on the 50 GB container disk

Problem	Effect	Fix
MADLAD with beam 4 and batch 32 ran out of GPU memory (24 GB)	wave-3 translation failed	automatic batch halving on out-of-memory
Parameter counter counted tied embeddings twice	Gemma 4.97B instead of 4.30B; Gemma+MADLAD wrongly reported as over 8B	count from the safetensors headers (Section 4.2)
Pod terminated instead of stopped (25 Oct)	the volume with the raw outputs of the follow-up runs was deleted	results copied off the pod before every shutdown; the most important runs repeated (Section 8.6)
Slow network on new pods (26 Oct: Python packages 0.05–1.2 MB/s, Hugging Face 1–2 MB/s per connection)	one pod unusable; setup slow	`runpod/speedtest.sh` before any install; only the final model downloaded
`runpodctl` transfers of a 4.7 MB archive arrived corrupted (same size, damaged middle)	code would not unpack	1 MB parts, each checked by MD5 and resent if needed

11 Reproducibility

Code. Package `sinmmlu/`: `data` (loading, normalisation, ids, permutations), `prompts` (prompt registry), `scoring` (averaging, calibration), `model` (loading, label/cloze scoring, generation), `evaluate` (runs conditions), `translate`, `ensemble`, `rationales`, `report` (paired statistics), `select_winner`, `predict_winner`, `submit` (format validation), `build_dev`, `dedup`, `tokens`, `debug`, and the analysis-only `negation` and `qtypes`. 38 offline tests (including a fake bigram model that checks batched label and cloze scoring against an unbatched reference, a simulated out-of-memory split, and stub models for self-consistency, the multi-turn negation chain and option-order averaging of the reasoning) run without a GPU. The code is in git (branch `initial-setup`, commit `f15939b`); reproducing submission 2 from a clean setup gives identical answers for all 2,238 test questions (Section 8.6). Every experiment stores its configuration, model revision, parameter count, library versions, GPU and git revision in `outputs/<experiment>/config.json`.

Commands (RunPod, after uploading the code and data):

```
export HF_TOKEN=...                       # read token; Gemma licence accepted
bash runpod/speedtest.sh                  # network check before installing anything
bash runpod/setup.sh                      # GPU check, pinned libraries, dev files, tests, gemma-3-4b-it only
bash runpod/rerun_1026.sh                 # reproduce submission 2 and compare; final system on the 1,483
bash runpod/run_subset.sh smoke           # then wave1, wave2, wave3, combos, confirm
bash runpod/select_and_test.sh            # selection rule -> submission 1
bash runpod/run_subset.sh cheap           # then translation, qtypes (follow-up screening)
bash runpod/confirm2.sh                   # second confirmation + selection -> submission 2
bash runpod/negchain_1026.sh              # final screening round (with runpod/last2_1026.sh)
# or directly, the final system only:
python -m sinmmlu.evaluate --model configs/models/gemma3_4b_it.yaml --data test \
       --prompt cot_en_role --chat on --max-new-tokens 1024
```

The submission is written to

`outputs/gemma3_4b_it__none__cot_en_role__chaton__p1__c0__trnone__mt1024__test/predictions.zip`

Greedy decoding makes the run deterministic up to floating-point differences between GPUs.

Cost. All runs together took about 13.5 GPU-hours on RTX 3090s (about 10 on 24 October, about 3.5 for the reproduction and the final screening round on 26 October): a label-scored condition takes 5–42 s for 280 questions (few-shot prompts are the slowest), a reasoning condition 6–9 minutes for 280 questions and 31–47 minutes for 1,483, and a test run 51 (512 tokens) to 71 minutes (1,024 tokens).

12 Limitations

- **All development data is Easy.** We have no labelled Medium or Hard questions, so the gains may not transfer to O/L and A/L questions, which are two thirds of the metric. In particular we cannot measure

whether the 140 answers that changed between submissions 1 and 2 (cut-off reasoning completed) are improvements; only the leaderboard shows it.

- **Single model, no training.** Fine-tuning (for example on Global-MMLU-si [5] and the public training data) was out of time; the paper rates the naturalness of Global-MMLU-si questions at 71/100 against 97/100 for native questions, so such data would need care.
- **Screening noise.** With 280 questions, gains below about 7 points cannot be separated; two subset results did not replicate. Only the confirmation results should be quoted as findings.
- **Parameter-count error** at selection time (corrected; no effect on the decision).
- **Process notes.** When submission 1 was checked, the counts of predicted answers (1–4) on the test set were printed once; for submission 2, the number of answers that differ from submission 1 was counted. Neither uses labels, nothing was chosen from them, and no other test-set statistic was computed.
- **Lost raw outputs.** The rented pod was terminated on 25 October, which deleted its volume before the raw outputs of the follow-up experiments (Section 8) had been copied. All numbers in this report were recorded in our lab notebook before that. On 26 October the most important runs were repeated with identical results (Section 8.6), restoring the per-question outputs of the final system on the confirmation set and its 2,238 test reasoning texts. Still missing are the per-question outputs of the other follow-up runs of 24 October, some per-type cells (Table 12) and the translation caches; greedy decoding and fixed seeds make them reproducible if needed. Raw outputs of the screening waves, combinations, the first confirmation, the final screening round and both submission files are kept.
- **Licences.** NLLB (used only in screening) is CC-BY-NC 4.0; the final system uses only Gemma (Gemma terms of use).

13 Next steps

1. Compare the leaderboard scores of submissions 1 and 2 (Medium/Hard effect of the longer reasoning limit).
2. Fine-tuning with LoRA on Global-MMLU-si (and the public training data, if the team agrees), judged by the same confirmation protocol, with **negation-focused training data** (for example "which is NOT …" questions and negated/non-negated question pairs) and a separate negation evaluation, since negation is the clearest failure mode of direct scoring and of translated input (Section 8.3); and data for the weakest types on the confirmation set: sub-option, fill-in-the-blank and numeric questions (Table 15).
3. The three-prompt reasoning ensemble (+1.1 on the confirmation set at 512 tokens, +1.8 on the subset at 1,024; not significant) if test-time compute and a larger labelled evaluation set allow a decision.

Explored and not pursued: self-consistency, translation pipelines with and without reasoning, option-by-option judgement, the negation-consistency chain, option-order averaging of the reasoning, calibration, cloze scoring.

References

[1] Ashmari Pramodya, Nirasha Nelki, Heshan Shalinda, Chamila Liyanage, Yusuke Sakai, Randil Pushpananda, Ruvan Weerasinghe, Hidetaka Kamigaito, and Taro Watanabe. 2025. SinhalaMMLU: A Comprehensive Benchmark for Evaluating Multitask Language Understanding in Sinhala. In *Proceedings of EMNLP 2025*.

[2] Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. 2021. Calibrate Before Use: Improving Few-Shot Performance of Language Models. In *Proceedings of ICML 2021*.

[3] NLLB Team. 2022. No Language Left Behind: Scaling Human-Centered Machine Translation. *arXiv:2207.04672*.

[4] Sneha Kudugunta et al. 2023. MADLAD-400: A Multilingual and Document-Level Large Audited Dataset. In *NeurIPS 2023 Datasets and Benchmarks*.

[5] Shivalika Singh et al. 2025. Global MMLU: Understanding and Addressing Cultural and Linguistic Biases in Multilingual Evaluation. In *Proceedings of ACL 2025*.

[6] Gemma Team. 2025. Gemma 3 Technical Report. *arXiv:2503.19786*.

[7] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *Proceedings of ICLR 2023*.

A Prompts (verbatim)

Shared pieces (Appendix B of [1]): head = "පහත දැක්වෙන්නේ බහුවරණ ප්‍රශ්නයකි." ("The following is a multiple choice question."); head-subject = "පහත දැක්වෙන්නේ {subject} සම්බන්ධ බහුවරණ ප්‍රශ්නයකි."; instr = "පහත දැක්වෙන ප්‍රශ්නය සඳහා නිවැරදි හෝ වඩාත්ම ගැළපෙන පිළිතුර 1, 2, 3, 4 පිළිතුරුවලින් තෝරන්න." ("Choose the correct or most suitable answer to the following question from the answers 1, 2, 3, 4."); body = "ප්‍රශ්නය: {question} / 1. … / 4. … / පිළිතුර:".

Name	Text (user turn; "/" = line break)
`paper_nosubject` (A0)	head / instr / / body
`paper_subject_si` (B1)	head-subject with the Sinhala subject name / instr / / body
`paper_subject_en` (B2)	as B1 with the English subject name
`strict_nosubject` (B3)	head / instr / "ඔබේ පිළිතුර ලෙස 1, 2, 3 හෝ 4 යන ඉලක්කම පමණක් ලියන්න. වෙනත් කිසිවක් ලියන්න එපා." ("Write only the digit 1, 2, 3 or 4 as your answer. Do not write anything else.") / / body
`role_subject_si` (B4)	"ඔබ {subject} විෂය පිළිබඳ ප්‍රවීණ ශ්‍රී ලාංකික ගුරුවරයෙකි." ("You are an expert Sri Lankan teacher of the subject {subject}.") / then B1
`en_instr_nosubject` (B5)	"The following is a multiple choice question in Sinhala. / Choose the correct or most suitable answer from the options 1, 2, 3, 4. / / Question: {Sinhala question} / options / Answer:"
`letters_nosubject` (B6)	as A0 with "A, B, C, D" in instr and options labelled A.–D.
`fewshot`k`_nosubject` (B7)	k earlier turns (A0 text of an example / "පිළිතුර: d"), then A0
`fewshot3_subject_si`	3 earlier turns in B1 format, then B1
`cloze_nosubject` (C4)	"පහත දැක්වෙන්නේ ප්‍රශ්නයකි. නිවැරදි පිළිතුර ලියන්න." ("The following is a question. Write the correct answer.") / / "ප්‍රශ්නය: {question} / පිළිතුර:" (no options)
`cot_si` (C6)	head / instr / "පිළිතුර තෝරා ගැනීමට පෙර පියවරෙන් පියවර සිංහලෙන් තර්ක කරන්න. අවසානයේ ``පිළිතුර: <අංකය>'' ලෙස ලියන්න." ("Before choosing the answer, reason step by step in Sinhala. At the end write 'Answer: <number>'.") / / question / options
`cot_en` (C7)	"The following is a multiple choice question in Sinhala. / Think step by step in English, then choose the correct or most suitable answer from the options 1, 2, 3, 4. / End your reply with "Answer: <number>". / / Question: {Sinhala question} / options"
`cot_en_subject`	as C7 with "…question in Sinhala about {English subject}."
`cot_en_role` (final)	"You are an expert Sri Lankan teacher of {English subject}." / then `cot_en_subject`
`cot_en_fewshot2`	2 earlier turns (C7 text of an example / the model's own correct English reasoning), then C7
`en_nosubject` (D1–D3)	"The following is a multiple choice question. / Choose the correct or most suitable answer from the options 1, 2, 3, 4. / / Question: {English translation} / translated options / Answer:"
`bilingual_nosubject` (D4)	"The following is a multiple choice question in Sinhala, followed by a machine translation into English. / The translation may contain errors; the Sinhala text is the original. / Choose …/ / Sinhala: / ප්‍රශ්නය: … / options / / English: / Question: … / translated options / Answer:"
`si_q_en_opts` (D5)	A0 with the options replaced by their English translations

B Example of a model-written worked example

Beginning of the English reasoning that Gemma wrote for an Arts pool question (gold answer 4), used as a worked example in `cot_en_fewshot2`:

```
Okay, let's break down this Sinhala multiple-choice question. The question describes a
statue or image of deities (likely Lakshmi and Saraswati) where one leg is bent and the
other leg is extended downwards. We need to identify the correct posture described.
...
Answer: 4
```

C Prediction distributions on the screening set

Share of predictions (%)	1	2	3	4
Gold answers	27.5	26.4	26.4	19.6
A0 baseline	22.9	30.0	33.6	13.6
4 option orders	26.1	26.4	26.1	21.4
Calibration	5.4	46.8	35.0	12.9
Gemma-3-4B-pt (base model)	46.4	35.7	11.8	6.1
`cot_en`	25.0	26.8	29.6	18.6
`cot_en_role`	26.4	26.8	26.8	20.0

Table 19: Predicted-option shares on the screening subset. Reasoning prompts give the most balanced predictions.

D Timeline

- 14–15 Oct: student phase: data checks (one label fixed), dedup, phase-0 evaluation of 0.6–1B models on a Colab T4, three scoring bugs found by review.
- 24 Oct: code rebuilt as the `sinmmlu` package; base model fixed to Gemma-3-4B-it; screening subset built; RunPod RTX 3090 set up; smoke test (pre-fill bug found and fixed); waves 1–3; combinations; confirmation; submission 1; parameter-count correction; code committed (`f11ed8e`); reasoning length, self-consistency and ensemble; translation with reasoning; negation and question-type analysis; second confirmation; submission 2.
- 25 Oct: code committed (`4a4aa89`); pod terminated (follow-up raw outputs lost, numbers kept).
- 26 Oct: one pod rejected after a network speed test; on a second pod (code `ae9abe5`) submission 2 reproduced exactly (2,238/2,238), final system rerun on the confirmation set; final screening round (negation chain, option-order averaging of the reasoning, 1,024-token ensemble); all results copied off the pod; code committed (`f15939b`).
- 27 Oct: leaderboard closes. 28 Oct: code. 30 Oct: report and slides.